THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The focal length of a concave lens is 1

(a) zero **(b)** positive **(c)** infinite **(d)** negative

2. The mirror used in a car's headlight is 1

(a) cylindrical **(b)** plane **(c)** concave **(d)** convex

3. A pencil kept in a glass of water looks bent due to 1

(a) scattering **(b)** refraction **(c)** dispersion **(d)** reflection

4. A convex lens of f = 100 cm has power 1

(a) +10 D **(b)** +0.1 D **(c)** +1 D **(d)** -1 D

5. The distance between the pole and focus is the 1

(a) aperture **(b)** image distance
(c) radius of curvature **(d)** focal length

6. The refractive index of glass is about 1

(a) 1.5 **(b)** 3.0 **(c)** 1.0 **(d)** 2.4

7. Magnification of a lens is 1

(a) f / v **(b)** u / f
(c) object size / image size **(d)** image size / object size

8. Radius of curvature is 1

(a) twice the focal length **(b)** equal to f
(c) four times f **(d)** half the focal length

9. Light bends away from the normal when it goes from 1

(a) glass to air **(b)** air to glass
(c) air to water **(d)** vacuum to diamond

10. A magnifying glass is a 1

(a) convex mirror **(b)** plane mirror **(c)** convex lens **(d)** concave lens

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the nature of a convex lens? 2

12. Define the radius of curvature of a mirror. 2

13. What is refractive index? 2

14. What are the signs of focal lengths of a convex lens and a concave lens? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A convex mirror has a radius of curvature of 30 cm. Find its focal length. 3

16. State the nature of the image formed by a convex lens when the object is placed at a large distance. 3

17. What is lateral displacement? Is the emergent ray parallel to the incident ray? 3

OR

State the three rules for drawing images in a concave mirror. 3

18. Give three differences between a convex lens and a concave lens. 3

OR

List three properties of the focal length of a concave lens. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain refraction through a rectangular glass slab. 5

OR

Compare the uses of concave and convex mirrors with examples. 5

20. Describe the images formed by a convex lens when the object is at 5
(a) 2F
(b) between F and O
(c) infinity.

OR

What are the images produced by a concave lens? Explain with a description of the ray diagram. 5

END OF QUESTION PAPER — Total: 40 Marks

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